MethylLlama: self-supervised foundation modeling of human DNA methylation profiles

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Abstract

DNA methylation is an informative biomarker of disease, tissue identity, and biological age, but conventional task-specific models do not fully exploit context-dependent regulation across the methylome or the large volume of public profiles lacking harmonized phenotype labels. Here we present MethylLlama, a self-supervised foundation model for human DNA methylation, built on a Transformer encoder and pretrained without phenotype supervision on 169,120 profiles from diverse tissues and studies, represented at 49,156 curated CpG sites. MethylLlama encodes CpG identity, quantitative methylation state, and genomic position. We adapt Whole-Cell Expression Decoding to methylation as the Whole-CpG Encoder–Decoder (WCED), which maps independently sampled partial views of each profile to global sample representations. Each representation is trained to reconstruct CpG values excluded from its input, while an InfoNCE objective aligns representations of the same sample across views. On held-out profiles, MethylLlama preserved sample identity across partial measurements and reconstructed withheld methylation values. Frozen representations captured age- and tissue-associated variation but also substantial study-specific structure. In chronological-age prediction, a five-model ensemble outperformed matched MethylGPT and ElasticNet pipelines on a common held-out cohort. These findings support MethylLlama as a reusable methylation representation model while identifying cross-study generalization as an important remaining challenge.

1 Introduction

DNA methylation is a covalent epigenetic modification that contributes to transcriptional regulation, chromatin organization, genome stability, and the maintenance of cellular identity. At CpG dinucleotides, methylation is commonly quantified as a continuous beta value using array-based assays that profile hundreds of thousands of loci [1, 2]. The biological meaning of a methylation measurement depends on both genomic and cellular context: regulatory CpGs can reflect altered gene activity, whereas tissue-specific methylation programs capture differentiation and cell composition [3, 4]. A methylation profile therefore represents an integrated molecular state rather than a collection of independent biomarkers.

The stability of DNA methylation, its accessibility across biological specimens, and its responsiveness to development, ageing, environmental exposure, and disease have made it a valuable substrate for biomarker discovery. Epigenetic clocks illustrate this potential. Early regularized models estimated chronological age across tissues [5], while later clocks were developed to capture mortality risk, health-related ageing, and the pace of physiological change [6–9]. These models remain strong and interpretable predictors, but each is optimized for a predefined endpoint; biological structure shared across methylation datasets must therefore be learned repeatedly for different phenotypes.

This task-specific paradigm is poorly matched to the scale and heterogeneity of public methylation data. Existing profiles span tissues, cohorts, disease states, and array platforms, while high-quality phenotype labels are available for only a subset of samples and differ among studies. Moreover, platform-defined CpG panels overlap only partially, so models must often operate on different subsets of the methylome. Conventional linear predictors perform well when the endpoint and CpG panel are fixed, but their predominantly additive formulation does not directly model context-dependent relationships among loci. Deep supervised models can capture such relationships, yet training a separate model for each application leaves the much larger unlabelled corpus unused and can generalize poorly from small labelled cohorts. These constraints motivate a foundation-model strategy that separates large-scale representation learning from task-specific supervision.

Self-supervised pretraining has enabled reusable representations in language, genomics, and single-cell transcriptomics [10–13]. In DNA methylation, MethylGPT established that a Transformer can be pretrained on a large collection of human methylomes and adapted to biological prediction tasks [14]. MethylGPT combines masked CpG prediction with whole-profile reconstruction from a global CLS representation, thereby encouraging that representation to retain sample-level information. Whole-profile decoding, however, does not explicitly require the representation of a sample to remain stable when different CpG subsets are observed, nor does it isolate recovery of loci that were entirely absent from the encoder input. These properties are important when a pretrained representation must transfer across incomplete profiles and non-identical measurement panels.

Here we present MethylLlama, a self-supervised foundation model for human DNA methylation. MethylLlama was trained from scratch, without phenotype labels, on 169,120 profiles represented at 49,156 curated CpG sites, using the large public methylation resource and CpG panel assembled for MethylGPT [14]. Each token combines CpG identity with its quantitative methylation value. A Transformer encoder integrates these measurements, while genomic-rank rotary position encoding preserves chromosomal order and a dedicated CLS token provides a global representation for downstream adaptation [15].

To learn this representation, we adapt the whole cell expression decoder objective introduced in BMFM-RNA [16] to methylation as the Whole-CpG Encoder–Decoder (WCED). Two independently sampled partial views are generated from each methylation profile and processed by a shared encoder. For each view, a decoder predicts the measured profile from the CLS bottleneck, but the reconstruction loss is evaluated only at valid CpGs excluded from that encoder input, so the objective cannot be satisfied by copying the input forward. A complementary response to the same problem is offered by scConcept, which discards feature reconstruction altogether and pretrains cell representations purely by contrasting disjoint gene-panel views [17]. We adopt that diagnosis but not that remedy: rather than replacing reconstruction, we retain it in bottleneck form and add an InfoNCE objective that aligns CLS representations of the two views from the same profile while distinguishing representations from different profiles [18]. WCED therefore applies two distinct pressures to the same embedding, which must both support recovery of unobserved methylation values and remain stable across CpG subsets.

We show that independently sampled CpG views preserve sample-specific identity and that the CLS bottleneck supports accurate reconstruction of CpGs withheld from the encoder. Without phenotype supervision, the pretrained representation captures substantial age- and tissue-associated variation, and fine-tuning for chronological-age prediction outperforms both a matched MethylGPT pipeline and an ElasticNet baseline on a common held-out cohort. At the same time, frozen representations encode strong study-specific structure, and transfer of age information to entirely unseen studies is heterogeneous. Together, these results establish MethylLlama as a reusable DNA methylation foundation model while identifying cohort heterogeneity as a central challenge for cross-study generalization.

2 Results

2.1 Study design and self-supervised pretraining

We pretrained MethylLlama on 169,120 human DNA methylation profiles represented in a shared vocabulary of 49,156 CpG sites. Pretraining was label-free: neither chronological age nor tissue, sex, dataset, or disease labels contributed to the objective. Each profile generated two independently sampled views, each containing 50% of its measured CpGs. A shared encoder compressed each view into a 256-dimensional CLS representation, from which a decoder predicted measured CpGs withheld from that view. A symmetric cross-view contrastive term encouraged the two CLS representations from the same profile to agree while retaining discrimination between profiles (Fig. 1).

We evaluated the pretrained encoder on the AltuMAge-derived collection, which contains 10,988 profiles and 21,368 measured CpGs before age-range filtering and duplicate removal. We used the original train, validation, and test partition (7,416/1,308/2,264 profiles) only for frozen representation diagnostics. The supervised age benchmark applied the prespecified age filter and methylation-fingerprint deduplication, generated five stratified training/validation folds, and retained a common fixed test set of 2,149 unique sample identifiers for every fold and both model families. This separation is important: the 2,264-profile split supports in-distribution probes, whereas all reported fine-tuning and paired-comparison metrics use the filtered and deduplicated 2,149-sample test set.

The pretraining corpus size was confirmed directly from the canonical data header (169,120 profiles $\times$ 49,156 CpG sites) and partitioned 80/10/10 by a fixed permutation, leaving 16,912 profiles that were never seen during pretraining and are used for the reconstruction analyses below. This count should not be conflated with the 154,063 post-QC figure reported for MethylGPT, whose curation pipeline differs.

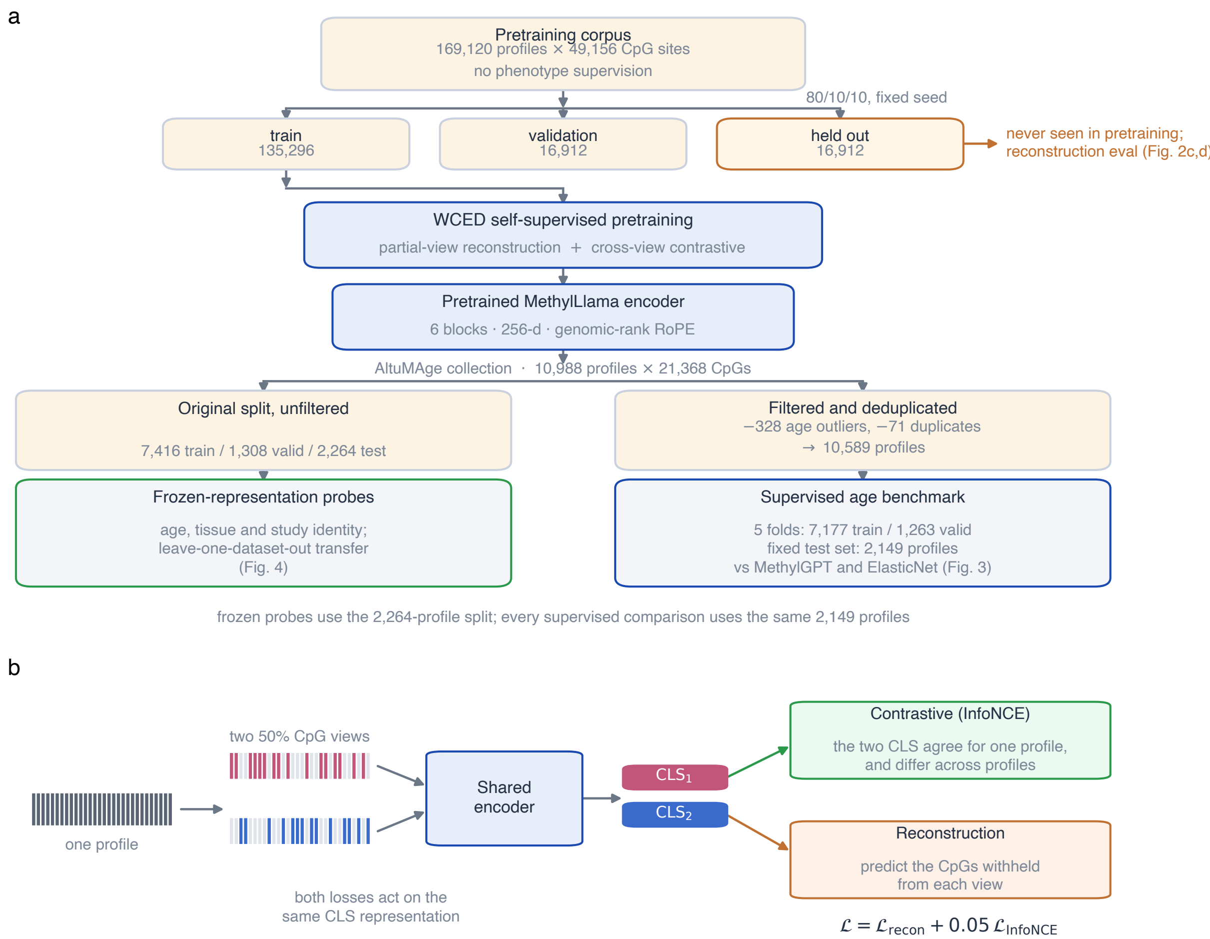

Figure 1: **Study design and self-supervised pretraining of MethylLlama.** **a**, Study design. The unlabelled pretraining corpus is distinguished from the labelled AltuMAge evaluation collection, and the two downstream analysis populations are shown separately: the 10,988-profile partition used for frozen-representation probes and the filtered, deduplicated 2,149-sample test set used for supervised comparisons. Phenotype annotations were not used as supervision during pretraining. **b**, View construction. Two independent 50% subsets of the measured CpGs of one profile form the two views; for each view, the complement (hatched) provides the reconstruction target, so the objective is scored only at measured CpGs absent from that view. **c**, The WCED objective. A shared encoder maps each view to a CLS representation; a decoder predicts the withheld CpGs from that representation alone, and a symmetric InfoNCE term aligns the two projected CLS embeddings. **d**, Token encoding. Each CpG token sums a learned site-identity embedding and a continuous β-value encoding, with a CLS token prepended to every partial profile.

2.2 WCED learns view-consistent global representations

WCED is designed to compress incomplete observations of a methylation profile into a shared global representation. Across five independent random draws of two half-profile views for each of 2,000 profiles, matched views had a mean cosine similarity of 0.984 ± 0.0004, compared with 0.499 ± 0.0002 for unmatched profiles, and the correct paired view was retrieved for 73.8% of queries (range 73.1–74.5% across draws; chance 0.05%). Retrieval reached 96.1% within the ten nearest candidates. The representation is therefore largely invariant to which half of the methylome is observed while remaining specific enough to identify the individual profile.

This invariance is nevertheless partly supported by shared input rather than by the representation alone. Because the two views are drawn independently, they share approximately half of their CpGs in expectation. Repeating the evaluation with strictly disjoint views—a random partition in which the two halves share no site—left mean matched similarity almost unchanged (0.971 ± 0.0005 versus 0.984) but reduced top-1 retrieval to 25.3% (range 24.6–26.0%). The dissociation is informative: average similarity is insensitive to the change, whereas identifying the correct partner among 1,999 competitors depends on narrow margins against the most similar profiles, which the small loss of positive similarity is sufficient to erase. Cross-subset identity information is clearly retained—disjoint-view retrieval remains roughly 500-fold above chance, and the correct partner falls within the ten nearest candidates for 75.1% of queries, comparable to top-1 performance under overlapping views—but it is weaker than the overlapping-view figure implies. We note that the encoder was trained with overlapping views, so disjoint evaluation is out of distribution and these values are a conservative bound rather than an estimate of what the architecture achieves when trained for that regime (Supplementary Fig. S1).

A related question is how far the representation of a partial profile departs from that of the complete profile, which bears on whether a sample assayed at fewer sites can be embedded consistently. The embedding of a random 50% CpG subset had a mean cosine similarity of 0.991 to the embedding of its own complete profile (shuffled-pairing control, 0.496), and a ridge age probe fitted on complete-profile embeddings and applied without adaptation to subset embeddings of held-out profiles retained $R^2 = 0.835$ against $R^2 = 0.863$ on complete profiles (paired difference 0.028, 95% CI 0.014–0.044; Supplementary Fig. S2). The representation is therefore substantially, though not perfectly, robust to which half of the measured sites is retained. We note that these are random subsets rather than the probe set of a particular array generation, so this does not by itself establish transfer across measurement platforms.

2.3 WCED reconstructs CpGs withheld from the input

We evaluated reconstruction on 5,000 profiles from the held-out portion of the pretraining corpus, encoding each from a 50% subset of its measured CpGs and scoring predictions only at measured CpGs absent from that input (1.12×10^8 evaluated positions). Predicted and observed beta values were closely correlated at these withheld sites (Pearson $r = 0.972$), and the per-profile standardized reconstruction loss was 0.0525—closely reproducing the checkpoint's own validation diagnostics

(0.9713 and 0.0552) on profiles excluded from pretraining (Fig. 2c).

To test whether the decoder genuinely reads the sample representation rather than exploiting population-level regularities, we compared three controls computed on the same profiles under identical withheld masks. Reconstruction error was 0.0056 for the model, against 0.0306 for a per-CpG mean predictor, 0.0505 for a decoder given standard normal vectors, and 0.0552 for a decoder given CLS embeddings shuffled across the batch (Fig. 2d). The model therefore reduced error 5.4-fold relative to the per-CpG mean, even though the mean predictor was computed on the evaluation cohort itself and is thus advantaged. Notably, shuffled CLS embeddings produced slightly worse reconstruction than random vectors: supplying a real but mismatched sample representation is more misleading than supplying no sample information at all. Together with cross-view consistency, this shows that the CLS bottleneck carries sample-specific information sufficient both to identify a profile and to predict methylation values never supplied to that view.

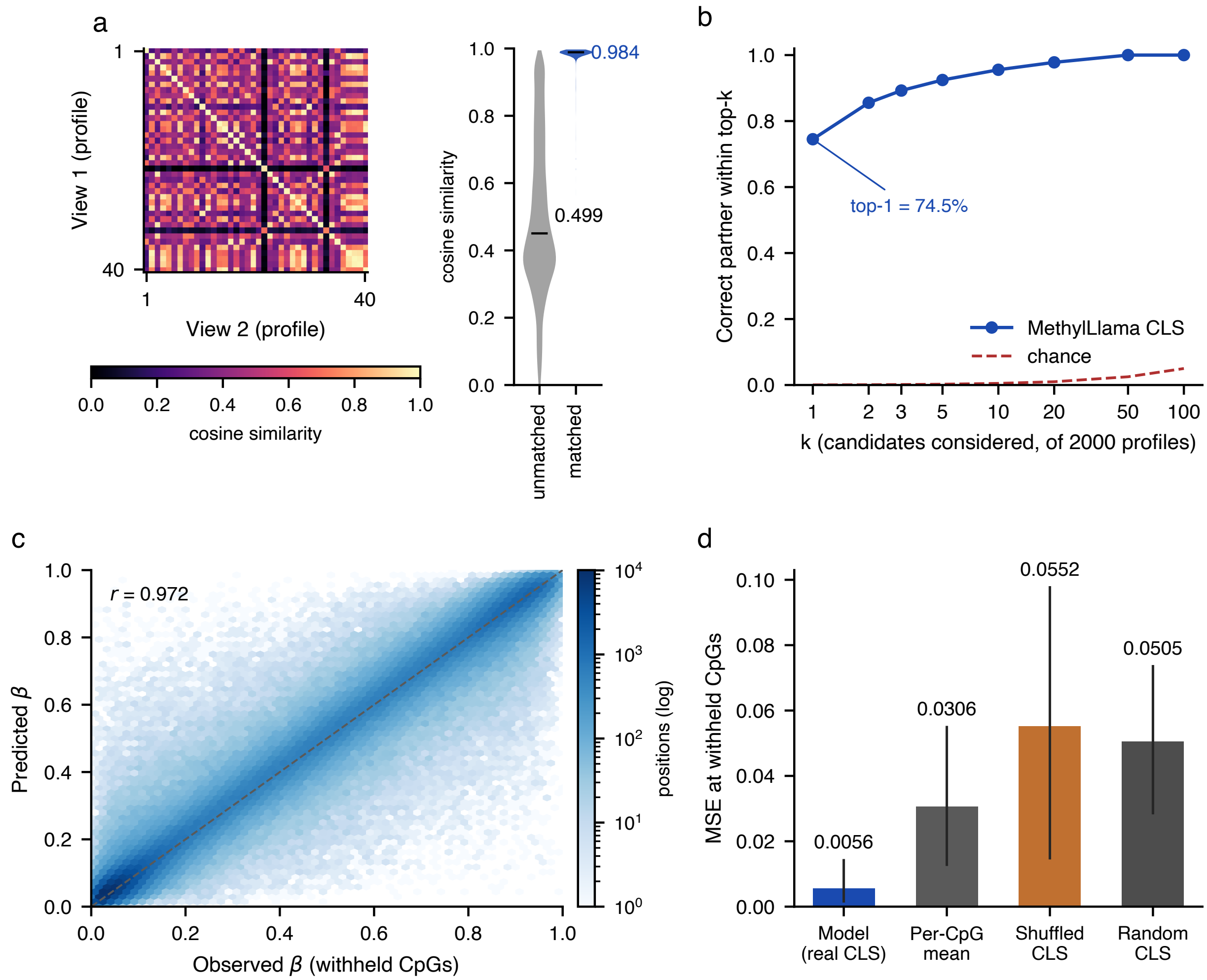

Figure 2: **WCED learns consistent partial-view representations and reconstructs withheld CpGs.** **a**, Cosine similarity between CLS embeddings of two independently sampled 50%-CpG views, for 40 representative profiles (heatmap; the bright diagonal marks matched views of the same profile) and summarised over all 2,000 profiles (violins; means annotated). **b**, Cross-view retrieval accuracy as a function of the number of candidates considered, against chance; the strictest point, top-1, is annotated. Panels a and b show one representative draw of two views per profile; values across five independent draws are given in the text and in Supplementary Fig. S1. **c**, Observed versus predicted β values at CpGs withheld from the encoder input, for 5,000 profiles from the held-out portion of the pretraining corpus (hexagonal binning, logarithmic colour scale; dashed line indicates identity). For legibility the panel displays a random subsample of 5.0×10^5 of the 1.12×10^8 evaluated positions; the reported correlation ($r = 0.972$) was computed over all evaluated positions, not over the plotted subsample. **d**, Mean squared error at withheld CpGs for the model and three controls evaluated on the same profiles under identical withheld masks. Bars show the mean across profiles and vertical lines span the 10th–90th percentile across profiles; these are dispersion ranges, not confidence intervals.

2.4 Pretraining captures age and tissue variation

We next asked whether self-supervised pretraining preserved information useful beyond its reconstruction objective. A ridge regressor fitted to frozen pretrained CLS embeddings on the original train-plus-validation profiles achieved $R^2 = 0.866$ and a median absolute error (MedAE) of 5.17 years on the original 2,264-profile test split. Applying the same probe to mean-pooled contextual CpG embeddings reduced performance to $R^2 = 0.795$ and MedAE 6.53 years. The CLS space also had a higher effective rank than the mean-pooled space (18.12 versus 3.40) and lower mean cosine anisotropy (0.500 versus 0.867). These findings indicate that the explicit sample token retained broader, more age-informative variation than uniform pooling across CpG tokens, despite receiving no age supervision during pretraining.

Biological information was not restricted to age. A linear classifier trained on frozen CLS embeddings attained balanced accuracy 0.936 across 37 sufficiently represented tissue classes (chance level 0.027). These probes establish that biologically associated structure is accessible in the pretrained representation under an in-distribution sample split. They do not show that the representation has separated biology from study or platform effects, because profiles from the same study can occur on both sides of the probe split.

An archived nonlinear replica-head probe reached $R^2 = 0.922$ and MedAE 3.57 years, but we treat it as exploratory: the current analysis script does not seed its NumPy minibatch shuffling and does not reserve a separate split for head-hyperparameter selection. The deterministic ridge probe is therefore the primary evidence for label-free age information.

2.5 Age prediction is stable across folds

We fine-tuned five models using different stratified training/validation folds and evaluated every selected checkpoint on the same 2,149-sample test set (age range 0–103 years; mean 45.53 years). Across folds, mean MedAE was 3.169 ± 0.040 years, mean MAE was 4.493 ± 0.059 years, and mean R^2 was 0.9316 ± 0.0024 (mean $\pm$ sample SD; Table 1). The narrow ranges across folds show that performance was not driven by one favorable supervised partition. Because the test samples were shared, however, these standard deviations measure training-partition stability and are not confidence intervals for population performance.

Averaging the five predictions for each sample produced the primary MethylLlama estimate, with MedAE 3.112 years, MAE 4.333 years, and $R^2 = 0.9365$. MethylGPT was aggregated in the same way for the paired neural-model comparison below.

Table 1: **Five-fold stability on the common held-out test set.** Values were recomputed from saved sample-level predictions. Because every model was evaluated on the same test samples, fold dispersion quantifies sensitivity to the training partition and is not a population confidence interval.

Model	MedAE (years)	MAE (years)	R^2
Fold 0	3.126	4.439	0.9321
Fold 1	3.213	4.458	0.9337
Fold 2	3.209	4.570	0.9284
Fold 3	3.152	4.541	0.9298
Fold 4	3.144	4.455	0.9338
Mean ± SD	3.169 ± 0.040	4.493 ± 0.059	0.9316 ± 0.0024
Fold ensemble	3.112	4.333	0.9365

2.6 MethylLlama improves age prediction over MethylGPT

We performed a paired comparison using identical held-out samples, age labels, and fold assignments for both model families. Fold-level predictions were averaged within sample, and uncertainty was estimated by resampling the same test samples for both models. MethylLlama reduced ensemble MedAE from 3.70 to 3.11 years (paired difference, 0.59 years; 95% CI, 0.34–0.81) and MAE from 5.11 to 4.33 years, while increasing R^2 from 0.911 to 0.936. Paired confidence intervals for all three endpoints excluded zero (Fig. 3).

We additionally evaluated a conventional linear reference using ElasticNetCV [19]. Hyperparameters were selected by internal cross-validation on the pooled training and validation profiles, with the common test set withheld until final evaluation. The resulting model used 1,036 non-zero coefficients among 21,368 CpGs and achieved MedAE 3.44 years, MAE 4.73 years, and $R^2 = 0.9322$. The MethylLlama ensemble therefore had better point estimates than both ElasticNet and MethylGPT on this test set. Because ElasticNet was evaluated as a single fitted model and its sample-level predictions were not retained, this comparison is descriptive and is not assigned a paired confidence interval.

The benchmark is matched at the sample and split levels, but the pipelines differ in their treatment of 1,760 CpGs outside the shared pretraining reference: MethylLlama retains their measured values under a common unknown-CpG identity, whereas MethylGPT excludes them. The result therefore establishes a predictive advantage for the implemented MethylLlama pipeline, but does not attribute that advantage to WCED alone.

Table 2: **Age-prediction performance on the common held-out test set.** The first three rows are test-set point estimates. MethylLlama and MethylGPT are five-fold prediction ensembles; ElasticNetCV is a single model selected by internal cross-validation. The final row reports the paired MethylGPT-minus-MethylLlama difference and its 95% bootstrap confidence interval; positive error differences and a negative R^2 difference favor MethylLlama.

Model or comparison	MedAE (years)	MAE (years)	R^2
MethylLlama	3.112	4.333	0.9365
ElasticNetCV	3.437	4.731	0.9322
MethylGPT	3.701	5.105	0.9110
MethylGPT − MethylLlama	0.589 [0.338, 0.815]	0.773 [0.583, 0.968]	−0.0254 [−0.0329, −0.0183]

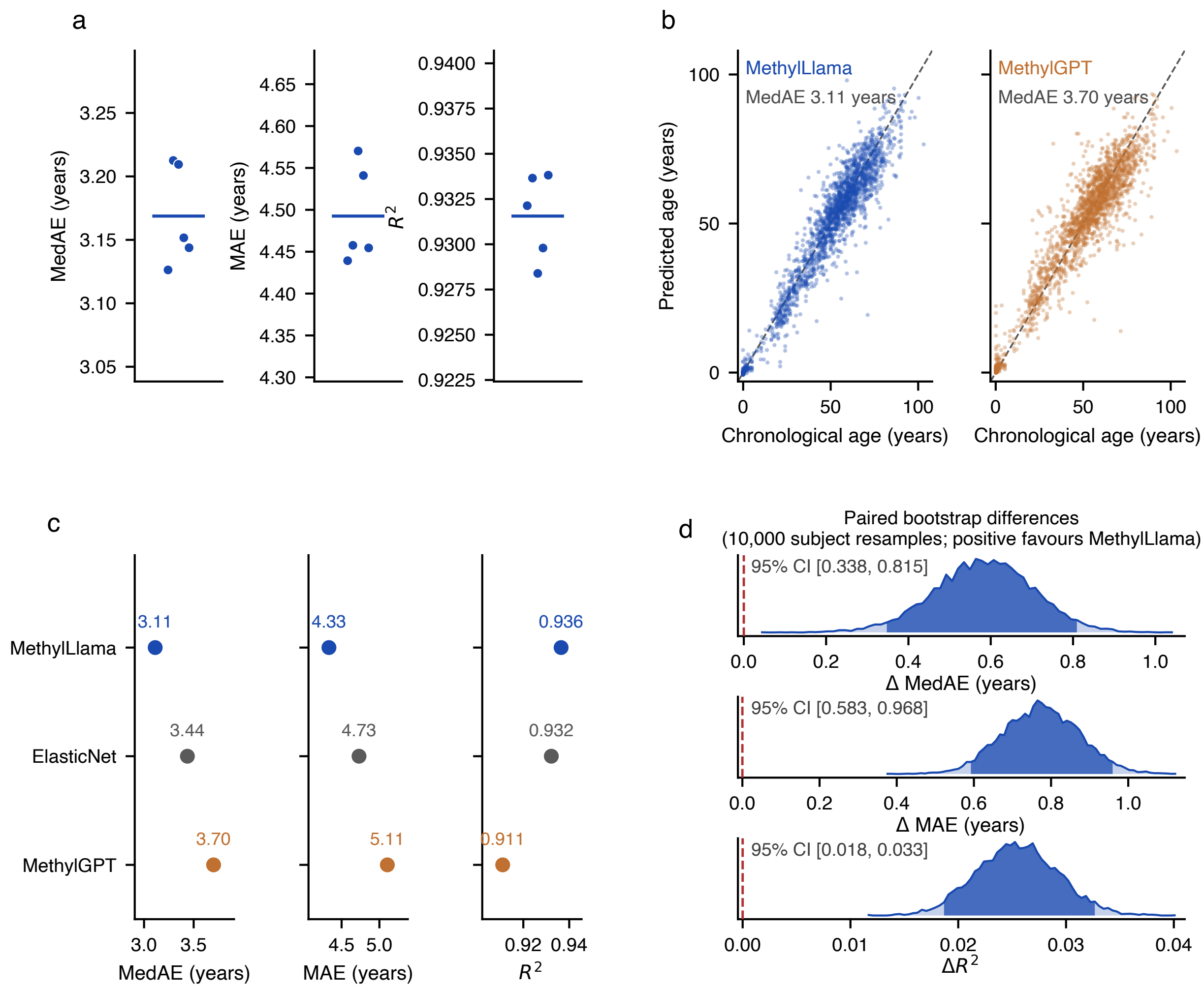

Figure 3: **Stable age prediction and comparison with neural and linear references. a**, Test-set MedAE, MAE, and R^2 for the five independently fine-tuned MethylLlama folds; horizontal lines denote fold means. **b**, Ensemble-predicted versus chronological age for MethylLlama and MethylGPT on the same 2,149 held-out samples; each point is one sample and dashed lines indicate identity. **c**, Ensemble point estimates for MethylLlama and MethylGPT and the single-model ElasticNetCV reference. **d**, Sampling distributions of the paired difference between models over 10,000 sample resamples, with the central 95% shaded; error reductions are MethylGPT-minus-MethylLlama differences, whereas the R^2 effect is MethylLlama-minus-MethylGPT, so positive values favor MethylLlama throughout. The vertical dashed lines mark no difference.

2.7 Genomic RoPE induces contextual CpG locality

To test the representational effect of genomic RoPE, we trained a matched pilot pair on the same fixed 5,000-profile subset. The two runs used the same architecture, objective, and hyperparameters; the intentional difference was whether RoPE received genomic-rank positions or sequential positions.

For the genomic-RoPE model, mean cosine similarity of contextualized representations was 0.9831 for adjacent CpGs and 0.8857 for random pairs, giving a locality gap of 0.0974. Without genomic positions, the corresponding values were 0.9931 and 0.9621, a gap of 0.0310. Thus, relative separation of adjacent from random CpGs was approximately 3.14-fold larger with genomic positions. In the production model's static token-embedding table, adjacent and random CpGs showed no positive locality gap, consistent with locality emerging through contextual computation rather than simple clustering of identity embeddings.

This ablation supports a mechanistic interpretation but remains a pilot. It used one seed, and the selected genomic checkpoint also had better reconstruction diagnostics than the non-genomic checkpoint (normalized reconstruction loss 0.0906 versus 0.1585; Pearson correlation 0.9357 versus 0.9017). We therefore report the locality result in the Supplementary Results and do not claim a downstream predictive advantage for genomic RoPE until replicated pretraining runs and a matched downstream ablation are available.

2.8 Pretrained representations retain tissue and study structure

Although tissue was highly predictable from frozen CLS embeddings, study identity was even more readily accessible: a linear probe attained balanced accuracy 0.956 across 85 eligible datasets, compared with a chance level of 0.012. Two-dimensional projections likewise formed many compact study-associated islands. This finding is not evidence of batch invariance. Instead, it shows that the representation simultaneously retains biological tissue structure and substantial cohort-specific structure. Tissue and dataset labels are also partially confounded in public methylation collections, so the tissue-probe accuracy cannot be interpreted as a study-independent biological metric.

For this reason, the main representation figure should pair the tissue result with the dataset result rather than show tissue alone. UMAP panels are descriptive and belong in the Supplementary Information; balanced-accuracy probes and cross-study tests provide the quantitative evidence (Fig. 4).

2.9 Cross-study transfer of age representations remains limited

We assessed whether age information in frozen CLS embeddings transferred to studies excluded from probe fitting. Whereas a random-split ridge probe achieved $R^2 = 0.856$, the median leave-one-dataset-out R^2 across 11 eligible studies was 0.261, with a median MedAE of 8.32 years; three studies yielded negative R^2. The broad variation between cohorts shows that readily decodable age information within a mixed-study split does not ensure robust transfer to an unseen study.

The failures are not explained by the obvious cohort descriptors. Neither the age dispersion of the held-out cohort (Spearman $\rho = +0.28$, $P = 0.40$) nor its sample size ($\rho = +0.26$, $P = 0.43$) was associated with transfer success, and the failing studies overlap the succeeding ones on both axes: one cohort of 118 profiles with an age SD of 10.5 years transferred at $R^2 = 0.42$, whereas a larger cohort of 282 profiles with a wider age SD of 13.9 years transferred at $R^2 = -4.63$ (Fig. 4c).

Cohort size and age range therefore do not account for the heterogeneity, which instead points toward study-specific biological or technical structure of the kind quantified above.

This is a diagnostic of the frozen representation, not a generalization test of the fully fine-tuned age model. Differences in tissue, disease status, platform, and age distribution also remain entangled with study identity. Claims of study-independent age prediction will therefore require grouped study-held-out fine-tuning with cohort-level uncertainty; the current analysis identifies cross-study transfer as an unresolved limitation.

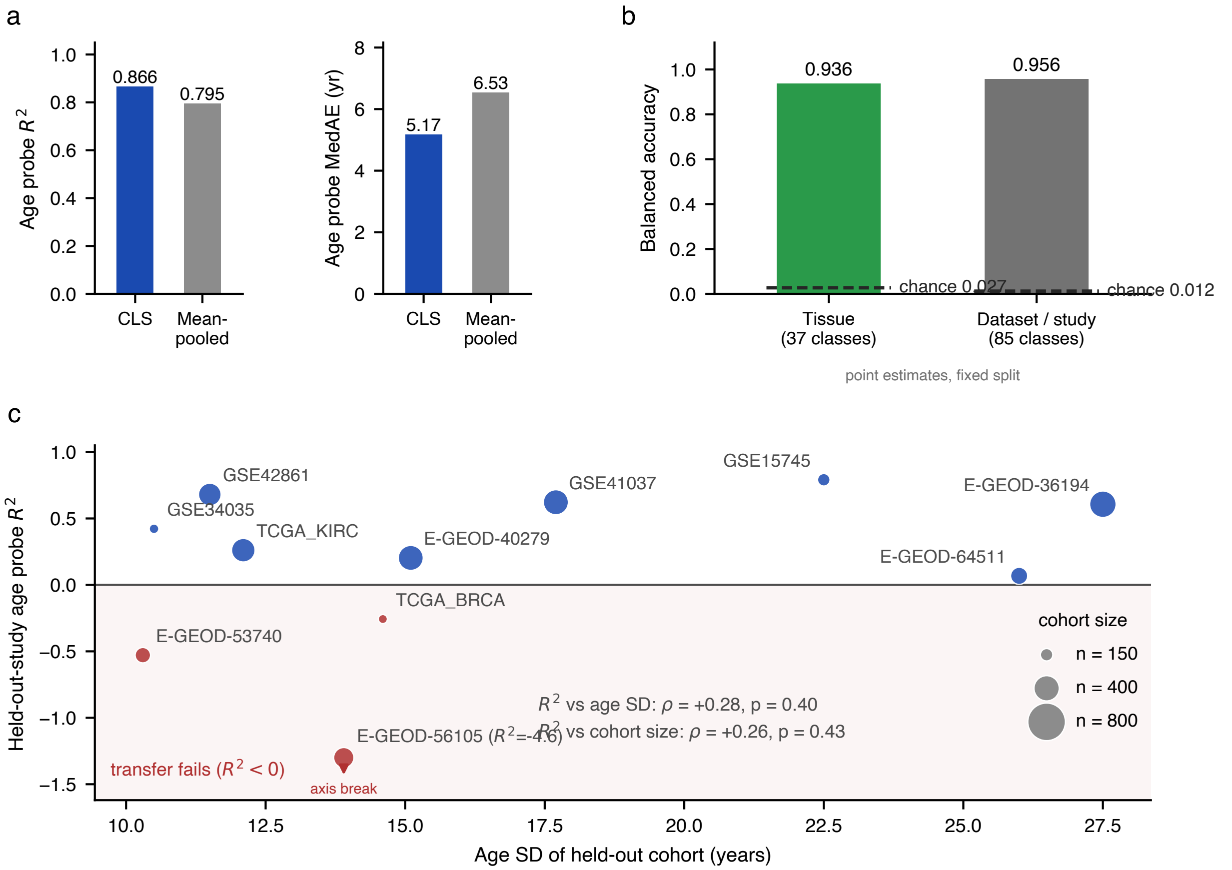

Figure 4: **Biological structure, study structure, and the limits of cross-study transfer.** **a**, Frozen linear age-probe performance using the pretrained CLS representation or mean-pooled contextual CpG embeddings. **b**, Balanced accuracy of frozen CLS linear probes for tissue and dataset/study identity; dark-grey dashed horizontal segments mark chance levels. **c**, Leave-one-dataset-out age-probe R^2 for the 11 studies meeting the prespecified size ($n \geq 100$) and age-dispersion (SD $\geq$ 10 years) thresholds, plotted against the age spread of the held-out cohort, with marker area proportional to cohort size. Studies with negative R^2 are shown in red on a shaded background; the most extreme value is clipped for readability and annotated with its true value. Neither cohort descriptor predicts transfer success (Spearman ρ inset). These analyses diagnose the frozen representation and do not constitute external validation of the fully fine-tuned model.

3 Discussion

We developed a partial-view self-supervised framework for learning global representations of DNA methylation samples. Unlike objectives that operate only at the level of masked features, WCED routes reconstruction through a CLS bottleneck and evaluates reconstruction at CpGs not provided to the encoder. Contrastive view matching directly trains the same sample embedding that is subsequently transferred to downstream analysis. The combination targets a distinction that is easy to miss in high-dimensional molecular modeling: successful prediction of masked features is not equivalent to learning a useful global representation.

The current results establish three narrow but meaningful points. First, the encoder can compress a partial methylation profile while retaining information about unobserved sites. Second, independently sampled views of the same profile remain identifiable in representation space. Third, the resulting encoder can be adapted reproducibly to age prediction. The CLS representation contained age-associated information before fine-tuning and retained more usable variation than mean pooling, consistent with the intended sample-level bottleneck.

The paired benchmark establishes better predictive performance for the implemented MethylLlama pipeline than for MethylGPT on the matched held-out samples. It does not, however, isolate the source of that difference: the pipelines vary in architecture, optimization, and their handling of 1,760 CpGs outside the shared pretraining reference. ElasticNet provided a strong non-neural reference, with performance close to the MethylLlama ensemble. Although all three point estimates favored MethylLlama, sample-level ElasticNet predictions were not retained, so statistical uncertainty is currently available only for the paired MethylGPT comparison. Matched objective and initialization ablations are still required before attributing the downstream gain specifically to WCED or to self-supervised pretraining.

The strongest limitation is cohort structure. Dataset identity is highly recoverable from the pretrained embeddings, showing that study- or batch-specific signals remain prominent. Random sample partitions can therefore overestimate transfer when samples from the same source study appear in model development and evaluation. This issue also changes how representation visualizations should be read. Separation by tissue may reflect genuine biology, but clean clusters can coexist with strong study effects. Quantitative study-held-out tests should carry more evidential weight than visually distinct UMAP clusters.

The representation analyses are therefore best viewed as diagnostics. Attention concentration, genomic locality, and CpG embedding structure can motivate hypotheses, but they do not by themselves identify causal age-associated loci or establish biological mechanism. Stronger interpretation would require stability across folds and seeds, perturbation-based importance, enrichment against a prespecified CpG background, and correction for multiple testing.

Further limitations include reliance on array-selected CpGs, incomplete evaluation across platforms, a single firmly established downstream phenotype, and no prospective clinical validation. A broad foundation-model claim would require transfer across multiple tasks and cohorts. At the present stage, “self-supervised methylation encoder” or “methylation representation model” more precisely describes the demonstrated scope.

In summary, the evidence supports WCED as a method for learning partial-view-consistent global methylation representations and supports MethylLlama as a stable age-prediction encoder on the held-out evaluation set. Whether the objective improves upon matched alternatives and whether the representation transfers to unseen studies remain the decisive experiments for the final manuscript.

4 Methods

4.1 Datasets and study design

Self-supervised pretraining used 169,120 public methylation profiles represented over 49,156 CpG sites. Profiles were partitioned once into training, validation, and held-out portions (80/10/10) by a fixed random permutation (seed 42); the held-out portion of 16,912 profiles was never seen during pretraining and provides the evaluation set for the reconstruction analyses reported below. The downstream age corpus was derived from the AltuMAge collection [20] and comprised 10,988 profiles over 21,368 CpG sites, spanning 37 tissue categories and 85 source studies. After removing 328 profiles with implausible age annotations (< 0 or > 120 years) and 71 duplicate profiles, 10,589 remained: 7,177 training, 1,263 validation, and a common held-out test set of 2,149 unique samples. Because the same test samples were evaluated by all five fold-trained models, variation across models is reported as training-fold stability rather than as an independent population confidence interval.

Split integrity. Partitions were defined at the level of individual profiles, indexed by their public sample accession; donor-level grouping was not available for the majority of source studies, which we note as a limitation. Duplicate and technical-replicate profiles were identified by methylation fingerprinting across all partitions, yielding 77 duplicate pairs spanning 136 profiles; within each connected component of the resulting graph the lexicographically first accession was retained and the remaining 71 profiles were excluded. The same exclusion list and age filter were applied identically to every model and baseline evaluated here, and set equality of the resulting 2,149 test accessions was verified programmatically rather than inferred from matching counts. Exact split manifests and their hashes will accompany the release.

Preprocessing. Beta values were restricted to $[0, 1]$. Pretraining used the full 49,156-CpG reference panel; downstream modelling used the 21,368 CpGs measured in the AltuMAge collection. Missing measurements were excluded from view sampling and from all loss terms through an explicit validity mask, so that no imputed value ever contributed to training or evaluation. CpGs were ordered by chromosome and genomic coordinate on the hg38 assembly using the Illumina HM450 manifest. No target-age labels were used during self-supervised pretraining: although the implementation exposes an optional auxiliary age-regression term, its weight was set to zero for the pretraining run reported here.

4.2 Input representation and encoder

Each observed CpG was represented by its site identity c_i and beta value b_i. The token embedding was

$$\boldsymbol{x}_i = s_c\, \boldsymbol{e}_{c_i} + f_\beta(b_i), \tag{1}$$

where $\boldsymbol{e}_{c_i}$ is a learned CpG embedding, s_c is a learned or initialized scale, and f_β is a continuous-value encoder based on sinusoidal basis functions followed by a linear projection. A CLS token was prepended to every partial profile.

The encoder used six transformer layers with hidden size 256, four attention heads (head dimension 64), feed-forward width 512, and a vocabulary of 49,161 entries (49,156 CpG sites plus five special tokens). Transformer blocks used pre-normalization with RMSNorm [21], multi-head self-attention [22], SwiGLU feed-forward layers, and rotary position embeddings [15, 23]. RoPE position identifiers were assigned using genomic rank rather than arbitrary input order. The pooled CLS output served as the sample representation.

4.3 Partial-view WCED pretraining

WCED adapts the Whole-Cell Expression Decoder introduced for transcriptomic foundation models [16] to array-derived DNA methylation. The original objective reconstructs a whole-cell expression profile through a CLS bottleneck; our formulation reconstructs CpG methylation values and evaluates reconstruction only at sites excluded from the encoder input.

The paired-view design was inspired by scConcept, which contrasts representations of different gene-panel views of the same cell [17]. For each methylation profile, two independently sampled views $V_i^{(1)}$ and $V_i^{(2)}$ contained 50% of valid CpGs. A shared encoder E_θ produced sample representations $\boldsymbol{h}_i^{(v)} = E_\theta(V_i^{(v)})_{\mathrm{CLS}}$. A shallow decoder D_ϕ mapped each CLS representation to a full-vector prediction $\hat{\boldsymbol{b}}_i^{(v)}$.

Reconstruction error was evaluated only over valid CpGs absent from the corresponding input view, denoted $H_i^{(v)}$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{2}\sum_{v=1}^{2}\frac{1}{|H_i^{(v)}|}\sum_{j\in H_i^{(v)}}\left(\tilde{b}_{ij} - \tilde{\hat{b}}_{ij}^{(v)}\right)^2, \tag{2}$$

where tildes denote per-sample standardisation: within each view, predicted and observed values at the withheld positions are centred and scaled to unit variance before the squared error is taken. This prevents the decoder from reducing the loss by predicting each CpG's marginal mean, and it makes the reported reconstruction loss scale-free.

A two-layer projection head g ($256 \to 128 \to 128$, ReLU) mapped CLS embeddings to L_2-normalized vectors $\boldsymbol{z}_i^{(v)}$. The symmetric cross-view InfoNCE loss used matched views as positives and other samples in the batch as negatives [18]:

$$\mathcal{L}_{\text{ctr}} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{\exp(\boldsymbol{z}_i^{(1)\top}\boldsymbol{z}_i^{(2)}/\tau)}{\sum_j \exp(\boldsymbol{z}_i^{(1)\top}\boldsymbol{z}_j^{(2)}/\tau)} + \log\frac{\exp(\boldsymbol{z}_i^{(2)\top}\boldsymbol{z}_i^{(1)}/\tau)}{\sum_j \exp(\boldsymbol{z}_i^{(2)\top}\boldsymbol{z}_j^{(1)}/\tau)}\right]. \tag{3}$$

The complete objective was $\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda\mathcal{L}_{\text{ctr}}$, with a fixed temperature $\tau = 0.1$ and $\lambda = 0.05$.

Relationship to contrastive single-cell pretraining. The contrastive component follows the cell-view formulation of scConcept [17], and we state the differences explicitly because they bound interpretation. First, and most importantly, contrastive alignment is the *sole* pretraining objective in that framework, which dispenses with reconstruction altogether; here it is an auxiliary term ($\lambda = 0.05$) accompanying withheld-CpG reconstruction. We adopt the underlying diagnosis—that feature-level reconstruction does not by itself optimise the sample-level embedding—but address it with two complementary mechanisms: routing reconstruction through the CLS bottleneck and scoring it only at withheld sites, and adding explicit cross-view alignment pressure on the same representation. Second, their two views are disjoint gene panels, whereas our two 50% CpG subsets are drawn independently and may overlap, which makes view alignment a comparatively easier task. Third, their negative set additionally contains embeddings from the anchor's own view, which discourages the encoder from representing panel identity rather than sample identity; our negative set contains only cross-view embeddings. Fourth, their temperature is learned during training whereas ours is fixed, and their contrastive loss is computed on the CLS embedding directly rather than through a projection head. Finally, their mini-batches are drawn from a single source dataset so that discrimination cannot be solved using inter-study technical differences, whereas ours are sampled at random from the pretraining corpus; we return to this point in the Discussion.

4.4 Age fine-tuning and comparators

A two-layer regression head ($256 \to 128 \to 1$, dropout 0.1) was applied to the CLS representation. The encoder was frozen for the first ten epochs while the head adapted, and then unfrozen and optimised jointly at a reduced encoder learning rate (2×10^{-5} versus 10^{-4} for the head). Fine-tuning used the complete measured profile as input rather than a partial view, and was optimised with a Huber loss on standardised age. Model selection used validation MedAE without access to the held-out test labels.

MethylGPT [14] was evaluated using the same source dataset, age labels, five fold assignments, and 2,149 held-out sample identifiers. Five predictions per sample were averaged within each model family before comparison. The pipelines differed in feature handling: 19,608 of the 21,368 downstream CpGs occurred in the common pretraining reference, whereas the remaining 1,760 CpGs were retained by MethylLlama under a shared unknown-site identity and excluded by the MethylGPT conversion pipeline.

As a non-neural reference, we fitted ElasticNetCV [19] to all 21,368 beta-value features after standardization based only on the pooled training and validation profiles. The documented l_1-ratio grid (0.1, 0.5, 0.7, 0.9, 0.95, 0.99, and 1.0), an automatically derived 100-value alpha path, and five-fold internal cross-validation were used once without post-hoc expansion of the search space. The selected model had $\alpha = 0.10883$, l_1 ratio 0.7, and 1,036 non-zero coefficients. The common test set was not used for hyperparameter selection. A single-fold random-initialization experiment was retained as exploratory and was not used for the primary comparison; matched objective ablations remain necessary to attribute downstream performance specifically to WCED.

4.5 Evaluation and statistical analysis

Primary endpoints were median absolute error (MedAE) and mean absolute error (MAE), measured in years. R^2 was secondary. Fold-level means and sample standard deviations quantify sensitivity to the training fold and are not population confidence intervals. For the MethylLlama–MethylGPT comparison, 95% confidence intervals were obtained from 10,000 paired percentile-bootstrap replicates with random seed 0. Held-out samples, rather than fold-level predictions, were resampled; the same bootstrap indices were applied to both model ensembles. The ElasticNet result is reported as a point-estimate reference because per-sample predictions were not retained for a paired analysis.

Representation analyses included matched-versus-unmatched view cosine similarity, cross-view retrieval, linear and nonlinear age probes, effective rank, and CLS-versus-mean pooling. All probe splits were isolated from probe fitting and hyperparameter selection.

Withheld-CpG reconstruction. Reconstruction was evaluated on 5,000 profiles drawn from the held-out portion of the pretraining corpus, which was excluded from pretraining by the fixed partition described above. Each profile was encoded from a 50% subset of its measured CpGs, and predictions were scored only at measured CpGs absent from that input—the same mask used by the training objective—giving 1.12×10^8 evaluated positions. Three controls were computed on the identical profiles under the identical masks: a per-CpG mean predictor, the decoder applied to CLS embeddings shuffled across the batch, and the decoder applied to standard normal vectors. The per-CpG means were computed on the evaluation cohort itself, which favours that baseline and therefore makes the comparison conservative.

Cross-view consistency. Two 50% views were encoded for each of 2,000 profiles and compared by cosine similarity of the resulting CLS embeddings. We report the mean similarity for matched and unmatched pairs, and retrieval accuracy at rank k: the fraction of first views whose matched second view falls within the k nearest of all 2,000 candidates. Retrieval at $k = 1$ is the strictest point on this curve; chance is $k/2000$. The whole evaluation was repeated for five independent random view draws, and we report the mean and range across draws rather than a single draw.

Two view constructions were compared. In the first, matching the construction used during pretraining, the two views are drawn independently and therefore share approximately half of their CpGs in expectation. In the second, they form a random partition of the valid CpGs and share no site, following the disjoint-panel construction of Bahrami et al. [17]. The disjoint condition is out of distribution for an encoder trained under the overlapping construction, and we therefore interpret it as a conservative bound on cross-subset consistency rather than as a matched comparison. Both conditions used the same profiles, checkpoint, and encoder call, so they differ only in how the two views are formed.

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